

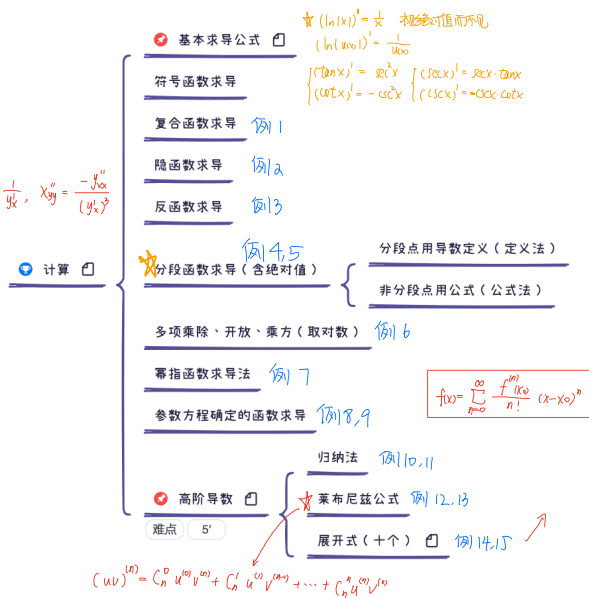
高等数学

难点: ① 计算量大 ② 含参数的讨论 ③ 高阶导数

第4讲 一元函数微分学的计算

$$\star x_{yy}'' = \frac{d(\frac{dy}{dx})}{dy} = \frac{d(\frac{dy}{dx})}{dx} \frac{dx}{dy} = \frac{-y''}{(y')^3}$$

$$x_y' = \frac{1}{y_x'}, x_{yy}'' = -\frac{y''}{(y_x')^3}$$

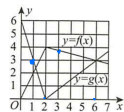


$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$

$$(uv)^{(n)} = C_n^0 u^{(n)} v^{(0)} + C_n^1 u^{(n-1)} v^{(1)} + \dots + C_n^k u^{(k)} v^{(n-k)}$$

例4.1

例 4.1 $f(x)$ 与 $g(x)$ 的图像如图 4-1 所示, 设 $u(x) = f[g(x)]$, 则 $u'(1) =$ _____.



$$u'(x) = f'(g(x)) \cdot g'(x)$$

$$u'(1) = f'(g(1)) \cdot g'(1) = f'(2) \cdot g'(1) = -\frac{1}{4} \cdot -3 = \frac{3}{4}$$

$$\frac{2}{\pi} \sin \frac{\pi}{2} t \Big|_1^y$$

例4.2

例 4.2 设 $y = y(x)$ 由方程 $x = \int_1^{y-x} \sin^2 \frac{\pi}{4} t dt$ 所确定, 则 $\frac{d^2 y}{dx^2} \Big|_{x=0} =$ _____.

$$y=1 \Leftrightarrow y-x=1 \quad 0 = \int_1^{y-x} \sin^2 \frac{\pi}{4} t dt = \frac{1}{2} \int_1^{y-x} (1 - \cos \frac{\pi}{2} t) dt = \frac{1}{2} [(y-x) - (\frac{2}{\pi} \sin \frac{\pi}{2} y - \frac{2}{\pi} \sin \frac{\pi}{2} 1)] \Rightarrow y-1 = \frac{2}{\pi} \sin \frac{\pi}{2} y - \frac{2}{\pi}$$

$$x' = 1 = [\sin^2 \frac{\pi}{4} (y-x)] \cdot (y' - 1)$$

$$y' = \frac{1}{\sin^2 \frac{\pi}{4} (y-x)} + 1 = \csc^2 \frac{\pi}{4} (y-x) + 1 \quad y'|_{x=0} = \csc^2 \frac{\pi}{4} + 1 = 3$$

$$\frac{d^2 y}{dx^2} = 2 \cdot \csc \frac{\pi}{4} (y-x) \cdot (\frac{\pi}{4} y' - \frac{\pi}{4}) (-\csc \frac{\pi}{4} (y-x) \cdot \cot \frac{\pi}{4} (y-x))$$

$$\sec x \rightarrow \sec x \tan x \\ \csc x \rightarrow -\csc x \cot x$$

$$\begin{aligned} \frac{d^2 y}{dx^2} \Big|_{x=0} &= 2 (\csc^2 \frac{\pi}{4} (y-x)) \cdot \cot \frac{\pi}{4} (y-x) \cdot \frac{\pi}{4} (1-y') \\ &= 2 \csc^2 \frac{\pi}{4} \cdot \cot \frac{\pi}{4} \cdot \frac{\pi}{4} (-2) \\ &= -\pi \cdot 2 \cdot 1 = -2\pi \end{aligned}$$

例4.3

例 4.3 设 $y = f(x)$ 与 $x = g(y)$ 互为反函数, $y = f(x)$ 可导, 且

$$f'(x) \neq 0, f(3) = 5, h(x) = f\left[\frac{1}{3}g^2(x^2+3x+1)\right], \text{ 求 } h'(1).$$

$$h(x) = f\left[\frac{1}{3}g^2(x^2+3x+1)\right]$$

$$h'(x) = f'\left(\frac{1}{3}g^2(x^2+3x+1)\right) \cdot \frac{2}{3}g(x^2+3x+1) \cdot g'(x^2+3x+1) \cdot (2x+3)$$

$$\begin{aligned} h'(1) &= f'\left(\frac{1}{3}g^2(5)\right) \cdot \frac{2}{3}g(5) \cdot 5 \cdot 3 \\ &= \underline{f'(3)} \cdot \underline{g'(1)} \cdot \frac{10}{3} \cdot 3 = 10 \end{aligned}$$

例4.4

例 4.4 设 $f(x)$ 在 $(-\infty, +\infty)$ 内连续且大于 0,

$$g(x) = \begin{cases} \frac{\int_0^x t f(t) dt}{\int_0^x f(t) dt}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

(1) 求 $g'(x)$;

(2) 证明: $g'(x)$ 在 $(-\infty, +\infty)$ 内连续.

$$(1) \quad x \neq 0 \text{ 时 } g'(x) = \frac{x f(x) \cdot \int_0^x f(t) dt - f(x) \int_0^x t f(t) dt}{\left[\int_0^x f(t) dt\right]^2}$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{x f(x)} = \lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{f(x)}$$



$$\lim_{x \rightarrow 0} \frac{g(x)-0}{x-0} = \lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{x \int_0^x f(t) dt} = \lim_{x \rightarrow 0} \frac{x f(x)}{\int_0^x f(t) dt + x f(x)} = \lim_{x \rightarrow 0} f(x) \cdot \frac{1}{f(x)} = 1$$

$$(2) \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} f(x) \cdot \frac{\int_0^x t f(t) dt + x f(x) - x f(x)}{2 \int_0^x f(t) dt \cdot f(x)} = \lim_{x \rightarrow 0} \frac{\int_0^x t f(t) dt}{2 \int_0^x f(t) dt} = \frac{1}{2} \neq 0$$

例 4.5

例 4.5 设 $f(x) = \begin{cases} x^\alpha \sin x^\beta, & x > 0, \\ 0, & x \leq 0, \end{cases}$ 其中 α, β 是常数, 且 $\beta < 0$. 问

- (1) 在什么情形下, $f(x)$ 不是连续函数?
- (2) 在什么情形下, $f(x)$ 连续但不可微?
- (3) 在什么情形下, $f(x)$ 可微, 但 $f'(x)$ 在 $[-1, 1]$ 上无界?
- (4) 在什么情形下, $f(x)$ 可微, 且 $f'(x)$ 在 $[-1, 1]$ 上有界, 但 $f'(x)$ 不连续?
- (5) 在什么情形下, $f'(x)$ 连续?

$$(1) \lim_{x \rightarrow 0^+} x^\alpha \sin x^\beta \neq 0 \Rightarrow \begin{cases} \alpha > 0, & \text{极限存在} \\ \alpha = 0, & \text{极限不存在} \\ \alpha < 0, & \end{cases} \quad \text{不是连续 } \alpha \leq 0.$$

$$(2) \text{连续 } \lim_{x \rightarrow 0} x^\alpha \sin x^\beta = 0 \Rightarrow \alpha > 0$$

$$\text{不可微: } \lim_{x \rightarrow 0^+} \frac{x^\alpha \sin x^\beta - 0}{x - 0} = \lim_{x \rightarrow 0^+} x^{\alpha-1} \sin x^\beta \neq 0 \Rightarrow \alpha - 1 \leq 0 \quad \left. \begin{array}{l} \alpha > 0 \\ \alpha - 1 \leq 0 \end{array} \right\} 0 < \alpha \leq 1$$

$$(3) \text{可微: } \alpha > 1$$

$$f(x) = \begin{cases} \alpha \cdot x^{\alpha-1} \sin x^\beta + x^\alpha \cos x^\beta \cdot \beta x^{\beta-1}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$f'(x) \text{ 无界: } \alpha + \beta - 1 < 0 \Rightarrow 1 < \alpha < 1 - \beta$$

$$(4) \text{有界: } \alpha > 1, \alpha + \beta - 1 \geq 0: \alpha \geq 1 - \beta$$

$$\lim_{x \rightarrow 0^+} \alpha x^{\alpha-1} \sin x^\beta + x^{\alpha+\beta-1} \cos x^\beta = \lim_{x \rightarrow 0^+} x^{\alpha+\beta-1} \cos x^\beta \neq 0 \Rightarrow \alpha = 1 - \beta$$

$$(5) f'(x) \text{ 连续: } \alpha > 1 - \beta$$

例 4.6

例 4.6 设 $y = \frac{x^3}{1-x} \cdot \sqrt[5]{\frac{2-x}{(2+x)^2}} + e^{4x}$, 求 y' .

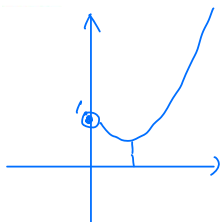
$$\ln|y| = 3\ln|x| - \ln|1-x| + \frac{1}{5}\ln|2-x| - \frac{2}{5}\ln|2+x|$$

$$\frac{y'}{y} = \frac{3}{x} - \frac{-1}{1-x} + \frac{1}{5} \frac{-1}{2-x} - \frac{2}{5} \frac{1}{2+x}$$

$$y' = \frac{x^3}{1-x} \cdot \sqrt[5]{\frac{2-x}{(2+x)^2}} \left(\frac{3}{x} + \frac{1}{1-x} + \frac{1}{5(2-x)} - \frac{2}{5} \frac{1}{2+x} \right) + 4e^{4x}$$

例 4.7

例 4.7 已知函数 $f(x) = \begin{cases} x^{2x}, & x > 0, \\ xe^x + 1, & x \leq 0, \end{cases}$ 求 $f'(x)$



$$f'(x) = \begin{cases} (e^{2x \ln x})' = x^{2x} \cdot (2 \ln x + 2), & x > 0 \\ e^x + xe^x, & x < 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} \frac{x^{2x} - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^{2x} - 1}{x} = \lim_{x \rightarrow 0^+} 2 \ln x = -\infty$$

例 4.8

$$q=1+2t^2 \quad t^2=4 \quad \ln t^2 = 2\ln t = \ln 4 \quad \ln t = \frac{\ln 4}{2}$$

例 4.8 设函数 $y = y(x)$ 由参数方程 $\begin{cases} x = 1 + 2t^2, \\ y = \int_1^{1+2\ln t} \frac{e^u}{u} du \end{cases} (t > 1)$ 确定, 求 $\left. \frac{d^2 y}{dx^2} \right|_{x=9}$.

$$\begin{aligned} \begin{cases} \frac{dx}{dt} = 4t \\ \frac{dy}{dt} = \frac{e^{1+2\ln t}}{1+2\ln t} \cdot \frac{2}{t} \end{cases} & \quad \frac{dy}{dx} = \frac{e^{1+2\ln t}}{1+2\ln t} \cdot \frac{1}{2t^2} = \frac{e}{2(1+2\ln t)} \\ \frac{d}{dt} \frac{dy}{dx} &= \frac{-e \cdot \frac{1}{t}}{4(1+2\ln t)^2} = \frac{-e}{4t(1+2\ln t)^2} \\ \left. \frac{d^2 y}{dx^2} \right|_{x=9} &= \frac{-e}{4t^2(1+2\ln t)^2} \bigg|_{t=2} = \frac{-e}{16(1+2\ln 2)^2} \end{aligned}$$

例 4.9

例 4.9 设函数 $y = y(x)$ 由 $\begin{cases} x = \arctan t, \\ 2y - ty^2 + e^t = 5 \end{cases}$ 确定, 求 $\left. \frac{dy}{dx} \right|_{t=0}$.

$$\begin{aligned} 2y + 1 = 5 & \Rightarrow y = 2 \\ \frac{dx}{dt} &= \frac{1}{1+t^2} \\ 2 \frac{dy}{dt} - y^2 - 2t \frac{dy}{dt} + e^t &= 0 \Rightarrow \frac{dy}{dt} = \frac{y^2 - e^t}{2(1-t)} \\ \left. \frac{dy}{dx} \right|_{t=0} &= \frac{\frac{4-1}{2}}{1} = \frac{3}{2} \end{aligned}$$

例 4.10

例 4.10 已知函数 $f(x) = \frac{x^2}{1-x^2}$, 求 $f^{(n)}(x) (n = 1, 2, 3, \dots)$.

$$\begin{aligned} f'(x) &= \frac{2x(1-x^2) - x^2(-2x)}{(1-x^2)^2} = \frac{2x}{(1-x^2)^2} \\ f''(x) &= -\frac{1-x^2-1}{1-x^2} = \frac{1}{1-x^2} - 1 = \frac{1}{(1-x)(1+x)} - 1 = \frac{\frac{1}{2}}{1-x} + \frac{\frac{1}{2}}{1+x} - 1 \\ f^{(n)}(x) &= \frac{(-1)^n n!}{2} [(1-x)^{-(n+1)} + (1+x)^{-(n+1)}] \end{aligned}$$

例 4.11

例 4.11 求 $y = x \ln x$ 的 n 阶导数, n 为大于 1 的正整数.

$$y' = \ln x + 1 \quad y'' = \frac{1}{x} \quad y^{(2+n)} = (-1)^n n! x^{-(n+1)}, n \geq 0$$



例4.12

例 4.12 设 $f(x) = \arctan x$, 求 $f^{(n)}(0)$.

[简单题: $f(x) = \arctan x$, 则 $f^{(6)}(0) = ? \rightarrow$ 递推性, 求导互换]

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(x) \cdot (1+x^2) = 1$$

$$u \cdot v = 1$$

$$f'(x) = u, (1+x^2) = v$$

$$(u \cdot v)^{(n-1)} = C_{n-1}^0 f^{(n)}(x) \cdot (1+x^2) + C_{n-1}^1 f^{(n-1)}(x) \cdot (2x) + C_{n-1}^2 f^{(n-2)}(x) \cdot 2 = 0$$

$$x=0: f^{(n)}(0) + (n-1) \cdot 2 f^{(n-2)}(0) = 0 \quad f^{(n)}(0) = -(n-1)(n-2) f^{(n-2)}(0)$$

$$\text{由 } f(0) = 0 \Rightarrow n \text{ 为偶数时 } f^{(n)}(0) = 0$$

$$f'(0) = 1 \Rightarrow n \text{ 为奇数时, } f^{(n)}(0) = (-1)^k (2k)! \quad n=2k+1$$

例4.13

例 4.13 设 $f(x) = (x^3 - 1)^n$, 则 $f^{(n)}(1) = \underline{\hspace{2cm}}$.

$$f(x) = [(x-1)(x^2+x+1)]^n = (x-1)^n (x^2+x+1)^n$$

$$f^{(n)}(1) = n! (1+1+1)^n = 3^n \cdot n!$$

例4.14

例 4.14 求函数 $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0 \end{cases}$ 在点 $x=0$ 处的 18 阶导数 $f^{(18)}(0)$.

$$\left\{ \begin{aligned} \frac{\sin x}{x} &= \frac{1}{x} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!} \\ \frac{\sin x}{x} &= \sum_{n=0}^{\infty} \frac{f^{(2n)}(0)}{(2n)!} x^{2n} \end{aligned} \right.$$

$$f^{(18)}(0) = 18! a_{18} = 18! \left(-\frac{1}{19!}\right) = -\frac{1}{19}$$

例4.15

例 4.15 设 $f(x) = (x-1)^5 e^{-x}$, 求 $f^{(10)}(1)$.

$$\textcircled{1} f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x-1)^n$$

$$\textcircled{2} f(x) = e^{-1} (x-1)^5 e^{-(x-1)} = \sum_{n=0}^{\infty} e^{-1} (x-1)^5 \cdot \frac{(-1)^n (x-1)^n}{n!} = \sum_{n=0}^{\infty} \frac{e^{-1} (-1)^n (x-1)^{n+5}}{n!}$$



$$\frac{f^{(10)}(1)(x-1)^{10}}{10!} = \frac{e^{-1}(-1)^5(x-1)^{10}}{5!} \Rightarrow -e^{-1} \frac{10!}{5!}$$

习4.1

4.1 设函数 $g(x)$ 可微, $h(x) = e^{1+g(x)}$, $h'(1) = 1$, $g'(1) = 2$, 则 $g(1) = (C)$.
(A) $\ln 3 - 1$ (B) $-\ln 3 - 1$ (C) $-\ln 2 - 1$ (D) $\ln 2 - 1$

$$h'(x) = e^{1+g(x)} \cdot g'(x) \Rightarrow h'(1) = e^{1+g(1)} \cdot 2 = 1 \Rightarrow e^{1+g(1)} = \frac{1}{2} \quad 1+g(1) = -\ln 2$$

习4.2

4.2 设函数 $f(x) = \int_{-1}^x \sqrt{1-e^t} dt$, 则 $y = f(x)$ 的反函数 $x = f^{-1}(y)$ 在 $y = 0$ 处的导数 $\frac{dx}{dy} \Big|_{y=0} = \frac{1}{\sqrt{1-e^{-1}}}$

$$f(x) = \int_{-1}^x \sqrt{1-e^t} dt \quad f'(x) = \sqrt{1-e^x}, \quad f'(-1) = \sqrt{1-e^{-1}}$$

习4.3

4.3 设函数 $y = y(x)$ 由参数方程 $\begin{cases} x = t + e^t \\ y = \sin t \end{cases}$ 确定, 则 $\frac{d^2 y}{dx^2} \Big|_{t=0} = -\frac{1}{8}$.

$$\begin{cases} \frac{dy}{dt} = \cos t \\ \frac{dx}{dt} = 1 + e^t \end{cases} \Rightarrow \frac{dy}{dx} = \frac{\cos t}{1 + e^t} = u$$

$$\frac{du}{dt} = \frac{-\sin t (1 + e^t) - \cos t \cdot e^t}{(1 + e^t)^2}$$

$$\frac{du}{dx} = \frac{-\sin t (1 + e^t) - \cos t \cdot e^t}{(1 + e^t)^3} \Big|_{t=0} = -\frac{1}{8}$$

习4.4

4.4 函数 $y = \ln(1-2x)$ 在 $x = 0$ 处的 n 阶导数 $y^{(n)}(0) = \frac{(-2)^n (n-1)!}{(1-2x)^{n-1}} \Big|_{x=0} = (-2)^n (n-1)! = -2^n (n-1)!$

$$y' = \frac{-2}{1-2x} \quad y^{(n)} = (-2)^n (-1)^{n-1} (n-1)! (1-2x)^{-n}$$

$$y^{(n)}(0) = (-2)^n (-1)^{n-1} (n-1)! = -2^n (n-1)!$$

习4.5

4.5 设 $y = y(x)$ 是由方程 $x^2 - y + 1 = e^y$ 所确定的隐函数, 则 $\frac{d^2 y}{dx^2} \Big|_{x=0} = 1$.

$$2x - \frac{dy}{dx} = e^y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2x}{1+e^y} \quad \frac{d^2 y}{dx^2} = \frac{2(1+e^y) - 2xe^y \cdot y'}{(1+e^y)^2} \Big|_{x=0} = \frac{2 \times 2}{4} = 1$$

习4.6

4.6 设 $y = \ln^3(\sin^2 x + 1)$, 求 y' .

$$y' = \frac{3 \ln^2(\sin^2 x + 1)}{\sin^2 x + 1} \cdot 2 \sin x \cos x$$

习4.7

4.7 设 $y = \arctan \frac{x-1}{x+1}$, 求 y' .

$$y' = \frac{\frac{(x+1) - (x-1)}{(x+1)^2}}{1 + \left(\frac{x-1}{x+1}\right)^2} = \frac{2}{(x+1)^2 + (x-1)^2} = \frac{1}{1+x^2}$$

4.8

4.8 设 $f(x) = (\cos x - 4)\sin x + 3x$.

(1) 求 $\frac{df(x)}{d(x^2)}$;

(2) 当 $x \rightarrow 0$ 时, $f(x)$ 为 x 的几阶无穷小?

$$(1) \frac{df(x)}{d(x^2)} = \frac{df(x)}{dx} \cdot \frac{dx}{d(x^2)} = \frac{-\sin^2 x + (\cos x - 4)\cos x + 3}{2x}$$

$$(2) \lim_{x \rightarrow 0} \frac{(\cos x - 4)\sin x + 3x}{x^5} = \lim_{x \rightarrow 0} \frac{4(1 - \cos x) - 2\sin^2 x}{x^4} = \lim_{x \rightarrow 0} \frac{4\left(\frac{x^2}{2} - \frac{x^4}{24} + o(x^4)\right) - 2\left(x^2 - \frac{x^4}{3} + o(x^4)\right)}{x^4} \neq A$$

5.1.1

$$\sin x = \left(x - \frac{x^3}{6}\right)^{\frac{1}{2}} = x^2 + \frac{x^6}{36} - \frac{x^4}{3}$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24}$$

4.9

4.9 设 $f'(0) = 1, f''(0) = 0$, 证明: 在 $x = 0$ 处, 有 $\frac{d^2}{dx^2} f(x^2) = \frac{d^2}{dx^2} f^2(x)$.

$$\left. \frac{d^2 f(x^2)}{dx^2} \right|_{x=0} = f''(x^2) \cdot 4x^2 + 2f'(x^2) \Big|_{x=0} = 0$$

$$\left. \frac{d^2 f^2(x)}{dx^2} \right|_{x=0} = \frac{d}{dx} (2f(x)f'(x)) \Big|_{x=0} = 2[f'(x)]^2 + 2f(x)f''(x) \Big|_{x=0} = 0$$

4.10

4.10 已知 $y = 1 + xe^{xy}$, 求 $y'(0), y''(0)$.

$$y = 1 + xe^{xy} \xrightarrow{x=0} y(0) = 1$$

$$y' = e^{xy} + xe^{xy}(y + xy') \xrightarrow{\substack{x=0 \\ y=1}} y'(0) = 1$$

$$y'' = 2e^{xy}(y + xy') + x[e^{xy}(y + xy')] \xrightarrow{x=0} y''(0) = 2$$

4.11

4.11 设 $f(x) = \begin{cases} x^2 e^{-x^2}, & |x| \leq 1, \\ \frac{1}{e}, & |x| > 1, \end{cases}$ 求 $f'(x)$.

$$f'(x) = \begin{cases} 2xe^{-x^2} + x^2 \cdot e^{-x^2} \cdot (-2x), & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

$$\lim_{x \rightarrow 1} \frac{x^2 e^{-x^2} - \frac{1}{e}}{x - 1} = \lim_{x \rightarrow 1} 2xe^{-x^2} - 2x^3 e^{-x^2} = 0$$

$$\lim_{x \rightarrow -1} \frac{x^2 e^{-x^2} - \frac{1}{e}}{x + 1} = \lim_{x \rightarrow -1} 2xe^{-x^2} - 2x^3 e^{-x^2} = 0$$

4.12

4.12 设 $y = [(1+x)(3+x)^9]^{\frac{1}{2}}(2+x)^4$, 求 $y'(0)$.

$$\ln|y| = \frac{1}{2}(\ln|1+x| + 9\ln|x+3|) + 4\ln|2+x|$$

$$\frac{y'}{y} = \frac{1}{2(1+x)} + \frac{9}{2(x+3)} + \frac{4}{2+x}$$

$$y'(0) = \left(\frac{1}{2} + \frac{3}{2} + 2\right) \left[3^9\right]^{\frac{1}{2}} 2^4 = 2^6 \cdot 3^{\frac{9}{2}}$$

习4.13

4.13 求 $y = \frac{5}{2+3x-2x^2}$ 的 n 阶导数, n 为正整数.

$$\begin{aligned} y(2+3x-2x^2) &= 5 \\ C_n^0 y^{(n)}(2+3x-2x^2) + C_n^1 y^{(n-1)}(3-4x) + C_n^2 y^{(n-2)}(-4) &= 0 \\ y &= \frac{5}{(2x+1)(2-x)} = \frac{+2}{2x+1} + \frac{-1}{x-2} = \frac{+2}{(2x+1)^{-1}} + \frac{-1}{(x-2)^{-1}} \\ y^{(n)} &= (-1)^1 (+2)^{n+1} n! (2x+1)^{-(n+1)} + (-1)^{n+1} n! (x-2)^{-(n+1)} \end{aligned}$$



习4.14

4.14 设 $y = (\arcsin x)^2$, n 为正整数, 证明:

$$(1-x^2)y^{(n+2)} - (2n+1)xy^{(n+1)} - n^2 y^{(n)} = 0,$$

并求 $y'(0), y''(0), \dots, y^{(n)}(0)$.

$$\begin{aligned} y &= (\arcsin x)^2 \\ y' &= 2 \arcsin x \cdot \frac{1}{\sqrt{1-x^2}} = 2 \arcsin x \cdot (1-x^2)^{-\frac{1}{2}} \Rightarrow \underline{2 \arcsin x = y' \sqrt{1-x^2}} \\ y'' &= 2(1-x^2)^{-\frac{3}{2}} + \arcsin x \cdot (1-x^2)^{-\frac{3}{2}} \cdot 2x \\ &= 2(1-x^2)^{-\frac{3}{2}} + y' \cdot x \cdot (1-x^2)^{-1} \\ \Rightarrow \underline{(1-x^2)y''} &= 2 + x \cdot y' \end{aligned}$$

$$\begin{aligned} C_n^0 y^{(n+2)}(1-x^2) + C_n^1 y^{(n+1)}(-2x) + C_n^2 y^{(n)}(-2) &= C_n^0 y^{(n+1)}x + C_n^1 y^{(n)} \\ y^{(n+2)}(1-x^2) - (2n+1)xy^{(n+1)} - n^2 y^{(n)} &= 0 \end{aligned}$$

$$\begin{aligned} x=0 \Rightarrow y_{(0)}^{(n+2)} &= n^2 y_{(0)}^{(n)} \\ y_{(0)}^{(1)} &= 0 \Rightarrow n=2k+1, y_{(0)}^{(2k+1)} = 0 \\ y_{(0)}^{(2)} &= 2 \Rightarrow n=2k, y_{(0)}^{(2k)} = [(k-1)!] \cdot 2^{(2k-1)} \end{aligned}$$

习4.1.1

1. 设函数 $y = f(x)$ 在 $x=0$ 的某邻域内二阶可导, $f(0) = 3, f'(0) = f''(0) = \frac{1}{2}$, 则

$$\left. \frac{d^2 x}{dy^2} \right|_{y=3} = (A).$$

(A) -4 (B) -2 (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

$$\begin{aligned} \frac{dx}{dy} &= 2 \\ \frac{d^2 x}{dy^2} &= \frac{d \frac{dx}{dy}}{dy} = \frac{d \frac{dx}{dy}}{dx} \frac{dx}{dy} = \frac{d \frac{1}{y_x}}{dx} \frac{1}{y_x} = \frac{-y_{xx}}{(y_x)^3} = \frac{-\frac{1}{2}}{\frac{1}{8}} = -4 \end{aligned}$$

4.1.2

2. 设函数 $f(x) = x^2 2^x$, 则对于任意正整数 $n > 1$, $f(x)$ 在 $x=0$ 处的 n 阶导数 $f^{(n)}(0) = (A)$.

- (A) $n(n-1)(\ln 2)^{n-2}$ (B) $n(n-2)(\ln 2)^{n-1}$
(C) $n(n+1)(\ln 2)^{n-2}$ (D) $n(n+2)(\ln 2)^{n-1}$

$$f(x) = C_n^0 x^2 \cdot 2^x \ln^0 2 + C_n^1 2x \cdot 2^x \ln^1 2 + C_n^2 2 \cdot 2^x \ln^2 2$$

$$= f^{(n)}(0) = n(n-1) \cdot \ln^2 2$$

4.1.3

3. 设 $f(x) = (x-1)^n x^{2n} \sin \frac{\pi}{2} x$, 则 $f^{(n)}(1) = (B)$.

- (A) $(n-1)!$ (B) $n!$
(C) $n! + 1$ (D) $(n+1)!$

$$\sin x = \frac{x^{2k+1}}{(2k+1)!}$$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(1)}{k!} (x-1)^k$$

$$f(x) = \sum_{k=0}^{\infty} \frac{(\frac{\pi}{2} x)^{2k+1}}{(2k+1)!} \cdot x^{2n} (x-1)^n$$

$$f(x) = [(x-1)^n] [x^{2n} \sin \frac{\pi}{2} x]$$

$$f^{(n)}(1) = C_n^0 n! \cdot 1 = n! \quad B$$

4.1.4

4. 设 $f(x) = xe^{-x}$, 则 $f^{(n)}(x) = (D)$.

- (A) $(-1)^n (1+n)xe^{-x}$ (B) $(-1)^n (1-n)xe^{-x}$
(C) $(-1)^n (x+n)e^{-x}$ (D) $(-1)^n (x-n)e^{-x}$

$$f^{(n)}(x) = x \cdot (-1)^n e^{-x} + n e^{-x} (-1)^{n-1} = (-1)^n e^{-x} (x-n)$$

4.1.5

5. 设 $f(x) = \begin{cases} x^3 \arctan \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases}$ 若 $f^{(n)}(x)$ 在 $x=0$ 处连续, 则 n 的最大值是 (B) .

- (A) 1 (B) 2 (C) 3 (D) 4

$$f'(x) = 3x^2 \quad f''(x) = 6x$$

4.1.6

6. 设函数 $f(x) = x^3 + 2x - 4$, $g(x) = f[f(x)]$, 则 $g'(0) = \underline{\quad\quad\quad}$.

$$f'(u) = 3u^2 \cdot u' + 2u' \quad \begin{cases} \Rightarrow g'(0) = 3 \times 16 \times 2 + 4 \\ = 100 \end{cases}$$

$$u(0) = -4 \quad u'(0) = 2$$

4.1.7

7. 设 $y = f(\frac{3x-2}{3x+2})$ 且 $f'(x) = \arctan x$, 则 $\frac{dy}{dx} \Big|_{x=0} = \underline{\quad\quad\quad}$.

$$y = f(\frac{3x-2}{3x+2}) \quad 6+6=12 \quad (\frac{3}{2}\pi)$$

$$\frac{dy}{dx} \Big|_{x=0} = f'(\frac{3x-2}{3x+2}) \cdot \frac{3(3x+2) - (3x-2)3}{(3x+2)^2} \Big|_{x=0}$$

$$= 3 \cdot \frac{2 \arctan(-1)}{1+1} = \frac{9}{2} \pi$$

$$= 3 \cdot \arctan(-1)^2 \cdot \frac{1}{1+1} = \frac{3}{4} \pi$$

千 4.1.8

$$\ln y = 0 \rightarrow y = 1$$

8. 设 $y = y(x)$ 由方程 $\ln(x^2 + y) = x^3 y + \sin x$ 确定, 则 $dy|_{x=0} = \underline{dx}$.

$$\frac{2x+y'}{x^2+y} = 3x^2 y + x^3 y' + \cos x$$

$$y' = 1$$

千 4.1.9

$$y = -y \Rightarrow y = 0$$

9. 已知可微函数 $y = y(x)$ 由方程 $y = -ye^x + 2e^y \sin x - 7x$ 所确定, 则 $y''(0) = \underline{-\frac{5}{2}}$.

$$y' = -y'e^x - ye^x + 2e^y(y' \sin x + \cos x) - 7$$

$$y'(0) = -y'(0) - y(0) + 2 - 7 \Rightarrow y'(0) = -\frac{5}{2}$$

$$y'' = -y''e^x - y'e^x - y'e^x - ye^x + 2e^y y' (y' \sin x + \cos x) + 2e^y (y'' \sin x + y' \cos x - \sin x)$$

$$2y''(0) = 2(-3) = -6 \Rightarrow y''(0) = -\frac{5}{2}$$

千 4.1.10

10. 设 $x = f(y)$ 是函数 $y = x + \ln x$ 的反函数, 则 $\frac{d^2 f}{dy^2} = \underline{\frac{x}{(x+1)^3}}$.

$$y'_x = 1 + \frac{1}{x} = \frac{x+1}{x} \quad y''_{xx} = -\frac{1}{x^2}$$

$$-\frac{y''_{xx}}{(y'_x)^3} = -\frac{\frac{1}{x^2}}{\left(\frac{x+1}{x}\right)^3} = \frac{x}{(x+1)^3}$$

千 4.1.11

11. 设 $y = \ln \sqrt{\frac{1-x}{1+x^2}}$, 则 $y''|_{x=0} = \underline{-\frac{3}{2}}$.

$$y = \frac{1}{2} (\ln(1-x) - \ln(1+x^2))$$

$$y' = \frac{1}{2} \left(\frac{-1}{1-x} - \frac{2x}{1+x^2} \right)$$

$$y''|_{x=0} = \frac{1}{2} \left(\frac{1}{(1-x)^2} - \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} \right) \Big|_{x=0} = \frac{1}{2} (-1 - 2) = -\frac{3}{2}$$

千 4.1.12

12. 设 $y = \frac{x^2}{1-x} \sqrt{\frac{2+x}{(2-x)^2}} + \sin x$, 则 $y' = \underline{\hspace{2cm}}$.

$$\ln|y| = 2\ln|x| - \ln|1-x| + \frac{1}{2} \ln|2+x| - \frac{2}{2} \ln|2-x|$$

$$\frac{y'}{y} = \frac{2}{x} + \frac{1}{1-x} + \frac{1}{2} \frac{1}{x+2} + \frac{2}{2} \frac{1}{2-x}$$

$$y' = \left(\frac{2}{x} + \frac{1}{1-x} + \frac{1}{2} \frac{1}{x+2} + \frac{2}{2} \frac{1}{2-x} \right) \frac{x^2}{1-x} \sqrt{\frac{2+x}{(2-x)^2}} + \cos x$$

4.1.13

13. 若 $f(x) = x^5 e^{6x}$, 则 $f^{(101)}(0) =$ _____.



$$f^{(101)}(x) = C_{101}^0 e^{6x} 6^{101} (x^5) + C_{101}^1 e^{6x} 6^{100} \cdot 5! (x^4) + C_{101}^2 e^{6x} 6^{98} \cdot 5! (x^3) + \dots$$

$$f^{(101)}(0) = \frac{101! \cdot 6^{96}}{96!} = \frac{6^{96} \cdot 101!}{96!}$$

4.1.14

14. 设 $f(x) = \frac{x}{1-2x^4}$, 则 $f^{(101)}(0) =$ _____.

$$f(x) \cdot (1-2x^4) = x$$

$$C_{101}^0 f^{(101)}(x) \cdot (1-2x^4) + C_{101}^4 f^{(97)}(x) \cdot (-48) = 0$$

$$f^{(101)}(0) = 48 C_{101}^4 f^{(97)}(0) = 2 \cdot 4! \cdot \frac{101!}{4! 97!} f^{(97)}(0) = 2 \cdot \frac{101!}{97!} f^{(97)}(0)$$

$$f'(x) = \frac{(1-2x^4) + 8x^4}{(1-2x^4)^2} = \frac{1+6x^4}{(1-2x^4)^2} \Big|_{x=0} = 1$$

$$f^{(101)}(0) = (48 \cdot \frac{101!}{4!})^{25}$$

4.1.15

15. 设 $f'(\ln x) = x \ln x$, 则 $f(x)$ 的 n 阶导数 $f^{(n)}(x) =$ _____.

$$f'(\ln x) = x \ln x$$

$$f'(x) = x e^x$$

$$f^{(n)}(x) = e^x x + (n-1) e^x$$

4.1.16

16. 设函数 $f(x)$ 二阶可导, $f'(0) = 1, f''(0) = 2$, 且 $\begin{cases} x = f(t) - \pi, \\ y = f(e^t - 1), \end{cases}$ 求 $\frac{dy}{dx} \Big|_{t=0}, \frac{d^2y}{dx^2} \Big|_{t=0}$.

$$\begin{aligned} x &= f(t) - \pi \\ y &= f(e^t - 1) \cdot 3e^{3t} \end{aligned}$$

$$\frac{dy}{dx} \Big|_{t=0} = \frac{f'(0) \cdot 3}{f'(0)} = 3$$

$$\frac{d^2y}{dx^2} \Big|_{t=0} = \frac{18 + 9 - 6}{[f'(0)]^3} = 2$$

4.1.17

17. 设函数 $y = y(x)$ 由参数方程 $\begin{cases} x = 1 + t^2, \\ y = \cos t \end{cases}$ 所确定, 求:

(1) $\frac{dy}{dx}$ 和 $\frac{d^2y}{dx^2}$;

(2) $\lim_{x \rightarrow 1^+} \frac{dy}{dx}$ 和 $\lim_{x \rightarrow 1^+} \frac{d^2y}{dx^2}$.

(1) $\frac{dy}{dt} = -\sin t \Rightarrow \frac{dy}{dx} = \frac{-\sin t}{2t}$

$$\frac{dx}{dt} = 2t \quad \frac{d^2y}{dt^2} = \frac{-\cos t \cdot 2t + \sin t \cdot 2}{4t^2} \quad \frac{d^2y}{dx^2} = \frac{2\sin t - 2t\cos t}{8t^3}$$

$$2) \lim_{x \rightarrow 1^+} \frac{-\sin t}{2t} = -\frac{1}{2}$$

$$\lim_{\substack{x \rightarrow 1^+ \\ t \rightarrow 0}} \frac{2\sin t - 2t\cos t}{8t^3} = \lim_{t \rightarrow 0} \frac{2\cos t - 2\cos t + 2t\sin t}{24t^2} = \lim_{t \rightarrow 0} \frac{2t}{24t^2} = \frac{1}{12}$$

4.1.18

18. 已知 $u = g(\sin y)$, 其中 $g'(v)$ 存在, $y = f(x)$ 由参数方程

$$\begin{cases} x = a \cos t, \\ y = b \sin t \end{cases} \quad (0 < t < \frac{\pi}{2}, a \neq 0, b \neq 0)$$

所确定, 求 du .

$$\begin{aligned} du &= g'(\sin y) \cdot \cos y \cdot y'_x \quad \frac{\cos t}{\sin t} = \frac{x}{y} \frac{b}{a} \\ &= g'(\sin y) \cdot \cos y \cdot \frac{b \cos t}{a \sin t} = -g'(\sin y) \cdot \cos y \cdot \frac{b^2}{a^2} \frac{x}{y} \end{aligned}$$

4.1.19

19. 设 $u = f[\varphi(x) + y^2]$, 其中 $y = y(x)$ 由方程 $y + e^y = x$ 确定, 且 $f(x), \varphi(x)$ 均有二阶导数,

求 $\frac{du}{dx}$ 和 $\frac{d^2u}{dx^2}$.

$$y' + e^y y' = 1 \Rightarrow y' = \frac{1}{1+e^y}$$

$$\begin{aligned} \frac{du}{dx} &= f'[\varphi(x) + y^2] \cdot (\varphi'(x) + 2y \cdot y') \\ &= f'[\varphi(x) + y^2] \cdot (\varphi'(x) + \frac{2y}{1+e^y}) \end{aligned}$$

$$\frac{d^2u}{dx^2} = f''[\varphi(x) + y^2] \cdot (\varphi'(x) + \frac{2y}{1+e^y})^2 + f'[\varphi(x) + y^2] \cdot (\varphi''(x) + \frac{2y}{(1+e^y)^3})$$

4.1.20

$$f(x) + 2f(x) = 3x$$

20. 设函数 $f(x)$ 满足 $f(x) + 2f(\frac{1}{x}) = \frac{3}{x} (x \neq 0)$, 求 $f'(x) (x \neq 0)$.

$$f'(x) - \frac{2f'(x)}{x^2} = -\frac{3}{x^2} \Rightarrow x^2 f'(x) - 2f'(x) = -3$$

$$-\frac{f'(x)}{x^2} + 2f'(x) = 3 \Rightarrow 4x^2 f'(x) - 2f'(x) = 6x^2$$

$$3x^2 f'(x) = 6x^2 + 3 \quad f'(x) = \frac{2x^2 + 1}{x^2} \quad (x \neq 0)$$

4.1.21

21. 设 n 为正整数, $f(x) = g'(x), g(x) = \begin{cases} \frac{e^x - 1}{x}, & x \neq 0, \\ 1, & x = 0, \end{cases}$ 求 $f^{(n)}(0)$.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\frac{e^x - 1}{x} - 1}{x - 0} &= \lim_{x \rightarrow 0} \frac{xe^x - (e^x - 1)}{x^2} = \lim_{x \rightarrow 0} \frac{xe^x}{2x} = \frac{1}{2} \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1 \\ \Rightarrow f(x) &= \frac{(x-1)e^x + 1}{x^2} \Rightarrow (x^2 f(x))^{(n+2)} = ((x-1)e^x + 1)^{(n+2)} \end{aligned}$$

$$x=0: (n+2)(n+1)f(x) = (n+2)e^x$$

$$f^{(n)}(0) = \frac{1}{n+1}$$

$$C_{n+2}^0 x^2 f^{(n+2)}(x) + C_{n+2}^1 2x f^{(n+1)}(x) + \underbrace{C_{n+2}^2 2 f^{(n)}(x)}$$

$$\underbrace{C_{n+2}^0 (x-1)e^x} + \underbrace{C_{n+2}^1 e^x}$$

$$(n+2)(n+1) f^{(n)}(0) = -e^x + (n+2)e^x = (n+1)e^x$$

$$f^{(n)}(0) = \frac{1}{n+2}$$